Managing Self-Learning Experts under Per-Round Budget Constraints

Ilgam Latypov MIPT
latypov.im@phystech.edu

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Abstract

This paper addresses the problem of sequential decision-making under learning budget constraints. Such settings naturally arise in applications like managing a portfolio of bandit or reinforcement learning (RL) algorithms. We propose a novel UCB-type algorithm, M-LCB, designed to manage a pool of K self-learning experts in a stochastic environment while accounting for a limited per-round learning budget M. At each round, M-LCB selects one expert to make a decision and at most $M \leq K$ experts to learn. For selection, M-LCB uses confidence bounds constructed from limited prior knowledge about the experts (i.e., mild assumptions) and their observed training losses. We derive anytime regret bounds for M-LCB that scale with the individual regrets of the experts. In particular, if each expert has regret $\tilde{O}(T^{\alpha})$ by round T, then M-LCB guarantees an overall regret of $\tilde{O}\left(\sqrt{KT/M} + (K/M)^{1-\alpha}T^{\alpha}\right)$ relative to the best expert in hindsight. Finally, we demonstrate the applicability of M-LCB using self-learning experts instantiated as (i) parametric models and (ii) bandit algorithms.

1 Introduction

In many applications, one must dynamically choose between multiple models: recommendation systems may run several predictors in parallel, updating them on incoming user feedback; financial platforms rely on switching between trading strategies as market regimes evolve; large-scale online services manage a portfolio of contextual bandits or reinforcement learning algorithms. These scenarios' objective is to dynamically select the most accurate model at each step, while managing a limited computational budget for training. This setup falls within the framework of sequential decision-making.

Classical multi-armed bandit (MAB) algorithms [Auer et al.(2002), Bubeck et al.(2012), Lattimore and Szepesvári(2020)], when addressing this problem, usually assume a static or adversarial reward distribution for each arm. Expert algorithms [Cesa-Bianchi and Lugosi(2006), Hazan et al.(2016)] usually require full feedback and do not account for how experts' learning rate. Neither approach fully addresses the challenge of managing multiple simultaneously-learning experts within a per-round training budget. We bridge this gap by proposing a procedure that unifies prediction with selective training, accounting for a fixed per-round computational budget. Specifically, the contributions of this work are as follows:

- **Novel UCB-Type Meta-Algorithm (M-LCB)**: we propose M-LCB, a novel Upper Confidence Bound (UCB)-type meta-algorithm. It manages a pool of K self-learning experts in a stochastic environment while accounting for a limited per-round learning budget $M (M \leq K)$.
- **Computational Efficiency**: we provide a method for constructing confidence bounds directly from realized losses. It is computationally efficient and sidesteps the need for expensive auxiliary optimization.

- **Theoretical analysis**: we estimate the meta-algorithm's performance in terms of the experts' individual convergence rates. For instance, when the experts' regrets are $\tilde{O}(n^\alpha)$, the overall regret scales as $\tilde{O}(\sqrt{KT/M} + (K/M)^{1-\alpha}T^\alpha)$.

- **Extension to Multi-Play Bandits:** we demonstrate that M-LCB extends to the multiple-play bandit setting and controls the top-M subset regret.

1.1 Related works

Self-learning experts (arms). The work [Dorn et al.(2025b)] introduces self-learning arms in the MAB setting: each arm is a black-box parametric function that generates a reward, and its parameter is updated after the arm is played. At each round, the learner selects an arm using a UCB-type index, observes the reward, and then updates the corresponding parameter.

Model selection at the meta-level. The work [Foster et al.(2017)] introduces a parameter-free aggregation of multiple online learners within the full information framework. CORRAL [Agarwal et al.(2017)] corrals a pool of bandit algorithms via log-barrier online-mirror descent (OMD) with importance-weighted feedback. The authors derive distribution-free guarantees in stochastic and adversarial settings. The work [Cutkosky et al.(2021)] proposes a dynamic balancing meta-algorithm based on known regret rate expressions for the base learners. In the setup, the regret is defined with respect to the globally optimal action, and only one learner is updated per round. In contrast, our formulation uses *per-expert, prefix-hindsight* guarantees $U_k(T, \delta)$ defined with respect to each expert's local optimum. The work [Dann et al.(2024)] removes the need for known regret rates by estimating per-learner coefficients online, obtaining high-probability, data-dependent model-selection guarantees for stochastic bandits (again, with a single learner updated per round). The closest setting to ours is [Pacchiano et al.(2020)]. It considers model selection using a *smoothing wrapper*. The authors show that the CORRAL meta-algorithm combined with their wrapper achieves regret $\tilde{O}(\sqrt{TK} + K^\alpha T^{1-\alpha} + K^{1-\alpha}T^\alpha c(\delta))$ when the regret of base learners satisfies $O(T^\alpha c(\delta))$. The dynamic balancing approach [Cutkosky et al.(2021)] yields a similar general bound $\tilde{O}\left(\sqrt{KT} + K^{1-\alpha}T^\alpha c(\delta)\right)$. These regret bounds coincide with the rate obtained by our method when only one expert can be trained per round. Our algorithm achieves the same order of dependence on T and α, while additionally supporting simultaneous training of up to M adaptive experts with confidence intervals computed directly from realized per-arm losses.

MABs with updates of multiple arms. In this setting, the meta-procedure can update or observe several arms per round. Several works consider the *adversarial* case. [Seldin et al.(2014)] study prediction with limited advice (query at most M arms). The authors obtain the regret bound $\tilde{O}\Big(\sqrt{\frac{KT\log K}{M}}\Big)$. It smoothly bridges the full-information case and the bandit setting. [Yun et al.(2018)] presents the minimax-optimal regret $\tilde{O}\Big(\max\{\sqrt{KT/M},\ \sqrt{T\log K}\}\Big)$. Specifically, it matches the lower bounds from [Seldin et al.(2014)]. However, these works assume non-learning arms (experts). In the *stochastic* case, [Yun et al.(2018)] considers the multi-armed bandit with additional observations: the learner plays one arm and may observe up to M extra arms per round. They propose the *KL-UCB-AO* algorithm that achieves asymptotically optimal *logarithmic* regret. However, it has a limited applicability scope due to the properties of the Kullback-Leibler-based selection rule.

Partial monitoring. Unlike standard partial monitoring [Cesa-Bianchi and Lugosi(2006)], where losses are estimated from signals, our model observes realized losses for selected experts. The key distinction is our Asm. 2, which assumes known improvement rates for experts and enables the use of confidence bounds. PM algorithms satisfying this assumption can nonetheless be incorporated as base experts in our framework.

Multiple-play multi-armed bandits. In the *multiple-play* setting (see [Agrawal et al.(1990), Uchiya et al.(2010)]), a meta-procedure selects M arms per round and observes semi-bandit feedback. UCB-based algorithms for combinatorial bandits [Kveton et al.(2015)] achieve $\tilde{O}(\sqrt{KT/M})$ regret under stochastic rewards, providing a baseline for subset-level performance analysis. These results serve as a benchmark for multiple-play MAB extension of M-LCB. Table 1 compares the problem setups.

Organization of the paper. Section 2 formalizes the setup. Section 3 presents the M-LCB algorithm. Section 4 develops the confidence-interval construction, the upper bounds, and the minimax lower bounds in one self-contained theoretical chapter, and Section 6 concludes the main narrative with the empirical study.

Table 1: Assuming $U_k(T,\delta) = O(T^\alpha c(\delta))$ with $\alpha \in [0,1]$ and poly-logarithmic $c(\delta)$, we compare regret rates up to logarithmic factors (✓ indicates supported properties). Regret follows Section 4.2.2 for multiple-play bandits, specific conventions for [Seldin et al.(2014), Yun et al.(2018)], and Section 2.2 otherwise. **Our M-LCB algorithm attains optimal rates for both regret definitions.**

Algorithm / Reference	Learnable	Multi-arm	Multiple-play	Regret rate (up to logs)
CORRAL + smoothing wrapper [Pacchiano et al.(2020)]	✓	×	×	$\tilde{O}\left(\sqrt{KT} + K^\alpha T^{1-\alpha} + K^{1-\alpha}T^\alpha c(\delta)\right)$
EXP3.P + smoothing wrapper [Pacchiano et al.(2020)]	✓	×	×	$\tilde{O}\left(\sqrt{KT} + K^{\frac{1-\alpha}{2-\alpha}} T^{\frac{1}{2-\alpha}} c(\delta)^{\frac{1}{2-\alpha}}\right)$
Dynamic Balancing [Cutkosky et al.(2021)]	✓	×	×	$\tilde{O}\left(\sqrt{KT} + K^{1-\alpha}T^\alpha c(\delta)\right)$
Prediction with Limited Advice [Seldin et al.(2014)]	×	✓	×	$\tilde{O}\left(\sqrt{\frac{KT\log K}{M}}\right)$
H-INF [Yun et al.(2018)]	×	✓	×	$\tilde{O}\left(\max\{\sqrt{KT/M},\ \sqrt{T\log K}\}\right)$
UCB-based multiple-play bandits [Kveton et al.(2015)]	×	×	✓	$\tilde{O}\left(\sqrt{KT/M}\right)$ $\alpha = \frac{1}{2}$
M-LCB [this work]	✓	✓	✓	$\tilde{O}\left(\sqrt{\frac{KT}{M}} + (K/M)^{1-\alpha}T^\alpha c(\delta)\right)$

2 Problem Setup

In this section, we formalize the interaction between the meta-procedure $\mathcal{P}$ and the pool of trainable experts. Section 2.1 details the interaction protocol. Section 2.2 defines the objective in terms of cumulative regret. Section 2.3 characterizes the learning dynamics of the experts, while Section 2.4 illustrates our framework through concrete examples in parametric learning and bandit selection.

2.1 Procedure protocol

We consider a stochastic online decision-making problem with K experts and a per-round training budget M. Let $\mathbf{U}$ be a decision space and $\mathbf{E}$ be an outcome space. The environment is characterized by a loss function $\ell : \mathbf{U} \times \mathbf{E} \to \mathbb{R}_+$. Each expert $k \in [K]$ operates on a specific subspace $\mathbf{U}_k \subseteq \mathbf{U}$ and generates a prediction $u_k^t \in \mathbf{U}_k$ at round t.

The meta-procedure $\mathcal{P}$ manages this pool of experts. At each round t, $\mathcal{P}$ allocates the training budget to a subset $S_t \subseteq [K]$ (where $|S_t| \leq M$) and selects a global decision $u^t \in \mathbf{U}$ by adopting the advice of a chosen expert. The environment then reveals a random outcome ξ^t taking values in $\mathbf{E}$, and $\mathcal{P}$ incurs the global loss $\ell(u^t, \xi^t)$.

Crucially, the expert learning is decoupled from the meta-procedure $\mathcal{P}$'s immediate performance. Experts update their internal parameters based solely on their own prediction loss, $\ell(u_k^t, \xi^t)$, and only when selected in the training set S_t. This creates a dual role for $\mathcal{P}$: it explores by allocating the training budget (choosing S_t to improve specific experts) and exploits by aggregating advice from the selected expert to form the final prediction u^t (see Section 3). This interaction mirrors the *prediction with limited advice* framework [Seldin et al.(2014)]. The box below summarizes the protocol.

Protocol: Decision making with trainable experts

For $t = 1, 2, \ldots$:

1. Choose $S_t \subseteq [K]$ with $|S_t| \leq M$ and decision $u^t \in \mathbf{U}$.
2. Observe $\xi^t \sim D$ and incur loss $\ell(u^t, \xi^t)$.
3. For each $k \in S_t$, compute $\ell(u_k^t, \xi^t)$ and update expert k.

2.2 Regret

Let $L(u) := \mathbb{E}_{\xi\sim\mathcal{D}}[\ell(u,\xi)]$ be the expected loss of a decision $u \in \mathbf{U}$. The optimal achievable expected loss $L_k^\star$ of k-th expert, and the global minimum $L^\star$ are,

$$L_k^\star := \min_{u\in\mathbf{U}_k} L(u), \qquad L^\star := \min_{k\in[K]} L_k^\star. \tag{1}$$

The goal of $\mathcal{P}$ is to minimize the cumulative regret relative to the best expert in the pool:

$$\mathrm{Reg}(T) := \sum_{t=1}^{T} \ell(u^t, \xi^t) - T \cdot L^\star.$$

This definition follows the standard regret formulation in the stochastic multi-armed bandit literature. We operate under the following standard assumptions on the environment:

Assumption 1 (Stochastic bounded losses)**.** *At each round, the environment draws $\xi^t \overset{\text{iid}}{\sim} D$ from an unknown distribution D supported on $\mathbf{E}$. The loss is $\ell : \mathbf{U} \times \mathbf{E} \to [0,1]$.*

Remark 1 (On the bounded loss)**.** *This setting considers bounded loss. However, it can be extended to the unbounded case (e.g., sub-Gaussian or heavy-tailed). Specifically, the proofs require a different choice of concentration inequalities. In this case, the regret guarantees hold up to a log-term.*

2.3 Self-learning experts

Each expert k is an online learning algorithm that improves as it is selected for training. Let $I_k(t) := \{\tau < t : k \in S_\tau\}$ denote the set of rounds up to time t where expert k was selected for training. We define $n_k^t := |I_k(t)|$ as the total number of such updates. We characterize the performance of these experts through an *anytime regret bound*. This ensures that as an expert is trained, its decisions u_k^t converge toward its optimum $L_k^\star$. The running loss and regret are

$$L_k(t) := \frac{1}{n_k^t} \sum_{\tau\in I_k(t)} \ell(u_k^\tau, \xi^\tau), \quad R_k(t) := n_k^t \left(L_k(t) - L_k^\star\right). \tag{2}$$

Assumption 2 (Anytime expert regret bound)**.** *For any $\delta \in (0,1)$, with probability at least $1-\delta$, for all $t \geq 1$, $R_k(t) \leq U_k(n_k^t, \delta)$, where $U_k(\cdot, \delta)$ is non-decreasing.*

Asm. 2 aligns with standard performance metrics in multi-armed bandits. We further show that regret in online optimization satisfies it.

2.4 Examples

Online parametric learning. Let $\xi^t = (x_t, y_t) \overset{\text{i.i.d.}}{\sim} D$, where the distribution D is supported on $\mathbf{E} := \mathcal{X} \times \mathcal{Y}$ and is unkown. The meta-procedure $\mathcal{P}$ aims to predict y_t observing x_t, i.e., it solves a supervised learning task. The decision space is a class of predictors, $\mathbf{U} := \{u : \mathcal{X} \to \mathcal{Y}\}$. $\mathcal{P}$ manages a pool of experts that are Online Convex Optimization (OCO) algorithms [Hazan et al.(2016)].

Each expert $k \in [K]$ is associated with a parameter space $\mathbf{W}_k \subseteq \mathbb{R}^{d_k}$, and a prediction function $f_k : \mathbf{W}_k \times \mathcal{X} \to \mathcal{Y}$. For any $w \in \mathbf{W}_k$, the mapping $f_k(w, \cdot)$ is a predictor in $\mathbf{U}$, so the decision space of this expert is $\mathbf{U}_k := \{f_k(w,\cdot) : w \in \mathbf{W}_k\} \subseteq \mathbf{U}$. The optimal risk for expert k (see (1)) turns into $L_k^\star = \min_{w\in\mathbf{W}_k} \mathbb{E}_{(x,y)\sim D}[\ell(f_k(w,x),y)]$.

When selected for training, expert k predicts $w_k^t \in \mathbf{W}_k$ and receives feedback $\ell(f_k(w_k^t, x_t), y_t)$. For example, if the expert is an Online Gradient Descent (OGD) optimizer, the loss $\ell(f_k(w_k,\cdot),\cdot)$ is assumed to be convex and G-Lipschitz on $\mathbf{W}_k$. The expert updates its parameters using the rule $w_k^{\tau+1} = \Pi_{\mathbf{W}_k}[w_k^\tau - \eta_\tau \nabla_w \ell(f_k(w_k^\tau, x_\tau), y_\tau)]$, where $\Pi_{\mathbf{W}_k}$ denotes the Euclidean projection onto $\mathbf{W}_k$, and η_τ is learning step size. The OCO regret characterizes the performance of such experts,

$$\overline{R}_k(t) := \sum_{\tau\in I_k(t)} \ell(u_k^\tau, \xi^\tau) - \min_{u\in\mathbf{U}_k} \sum_{\tau\in I_k(t)} \ell(u, \xi^\tau). \tag{3}$$

For OGD, it holds $\overline{R}_k(t) \leq O(GD_k\sqrt{n})$, where $D_k := \mathrm{diam}(\mathbf{W}_k)$ and t is s.t. $|I_k(t)| = n$ [Hazan et al.(2016)].

Lemma 2 relates this regret to the Assumption 2 with additional concentration term. Consequently, OGD experts satisfy the assumption with $U_k(n,\delta) = O\left(GD_k\sqrt{n} + \sqrt{n\log(n/\delta)}\right)$.

Multi-armed bandit algorithms. Motivated by AutoML and distributed routing, we consider a meta-procedure $\mathcal{P}$ that selects the most efficient among K self-optimizing experts. As in Bandit Model Selection [Pacchiano et al.(2020)], this is inherently challenging due to the experts' non-stationary learning dynamics and the high sample cost required for convergence.

Each expert k is a stochastic bandit managing a finite action space $\mathcal{A}_k := \{a_k^1, \dots, a_k^{m_k}\}$. An expert's decision at time t is a probability distribution $u_k^t \in \Delta_k$ over its actions, where $\mathbf{U}_k = \Delta_k$ is the m_k-dimensional simplex. The global decision space $\mathbf{U}$ is the union of simplices.

The environment is characterized by a latent random variable $\xi^t \sim D$ that represents the state of the system at time t (e.g., a vector of current latencies or resource costs for all available methods). Although ξ^t need not be observed in full by the meta-procedure, it determines the loss of each action $a \in \bigcup_k \mathcal{A}_k$ via the mapping $\tilde{\ell}(a, \xi)$. The expert loss for a decision $u_k \in \mathbf{U}_k$ is $\ell(u_k, \xi) := \mathbb{E}_{a \sim u_k}[\tilde{\ell}(a, \xi)]$. The optimal risk for expert k is $L_k^\star = \min_{a \in \mathcal{A}_k} \mathbb{E}_{\xi \sim D}[\tilde{\ell}(a, \xi)]$.

When expert $k \in S_t$ produces u_k^t, it samples a discrete action $a_k^t \in \mathcal{A}_k$ from u_k^t and receives $(a_k^t, \tilde{\ell}(a_k^t, \xi^t))$ as bandit feedback. Thus, in this instantiation the limited-advice assumption means that the environment provides the sampled-action loss for each trained expert $k \in S_t$, including experts that are not deployed as the global advisor. If several trained experts sample the same action, the corresponding loss need only be evaluated once and can be reused for all of them. When off-policy updates are available, one may also update base algorithms from additional samples, but our analysis uses only the on-policy observations above.

In this setting, the expert's performance is measured by its pseudo-regret $R_k(n)$, which evaluates the cumulative expected loss of the distributions u_k^τ relative to the best fixed action, $R_k(t) = \sum_{\tau \in I_k(t)} \mathbb{E}_{a \sim u_k^\tau}[\tilde{\ell}(a, \xi^\tau)] - n_k L_k^\star$.

Standard algorithms such as UCB satisfy Asm. 2 with $U_k(n, \delta) = O(\sqrt{m_k n \log(n/\delta)})$. The theoretical running loss $L_k(t)$ in (2) is defined through the conditional loss $\ell(u_k^t, \xi^t)$, whereas the bandit expert observes the sampled loss $\tilde{\ell}(a_k^t, \xi^t)$. In implementation one therefore uses $\tilde{L}_k(t) = \frac{1}{n_k^t} \sum_{\tau \in I_k(t)} \tilde{\ell}(a_k^\tau, \xi^\tau)$, and adds the corresponding sampling-concentration term to U_k. The concentration of $\sum_{\tau \in I_k(t)} \tilde{\ell}(a_k^\tau, \xi^\tau)$ around $\sum_{\tau \in I_k(t)} \ell(u_k^\tau, \xi^\tau)$ ensures that anytime high-probability bounds on pseudo-regret hold. The proof is similar to Lemma 2.

3 The M-LCB algorithm

We balance optimistic learning with conservative deployment: we allocate the training budget to potentially optimal experts with the lowest lower confidence bounds ($\texttt{LCB}_k$), while selecting the deployed advisor using the lowest upper confidence bounds ($\texttt{UCB}_k$) to minimize immediate loss. This prevents over-exploiting uncertain experts.

Definition 1 (UCB and LCB)**.** *Fix $k \in [K]$ and let*

$$\mathit{LCB}_k(t, \delta) := L_k(t) - \frac{U_k\left(n_k^t, \delta_{\mathrm{arm}}\right)}{n_k^t}, \quad \mathit{UCB}_k(t, \delta) := L_k(t) + H(n_k^t, \delta_{n_k^t}),$$

where $\delta_{\mathrm{arm}} = \frac{\delta}{2K}$, $\delta_n = \frac{\delta}{4Kn^2}$ and $H(n, \delta) = \sqrt{\frac{2\log(1/\delta)}{n}}$, with $U_k(\cdot)$ being the regret bound from Asm. 2 and $L_k(t)$ being an accumulated loss defined in (2)*.*

At round t, we compute $\texttt{LCB}_k(t, \delta)$ and $\texttt{UCB}_k(t, \delta)$ for each expert k. We then determine the training subset S_t and the advisor i_t using the following rules:

$$S_t := \arg\min_{\substack{S \subseteq [K] \\ |S| = M}} \sum_{k \in S} \texttt{LCB}_k(t, \delta), \quad i_t = \arg\min_{k \in S_t} \texttt{UCB}_k(t, \delta).$$

To address the lack of pointwise convergence in stochastic algorithms [Cesa-Bianchi and Lugosi(2006), Shalev-Shwartz et al.(2012), Pacchiano et al.(2020)], we use smoothing wrappers [Orabona(2019), Pacchiano et al.(2020)] to robustly construct the decision u^t of expert i_t, ensuring $\mathcal{P}$ satisfies Asm. 3. Unlike adversarial aggregation (even when $M = K$), our confidence bounds estimate optimal values by directly exploiting the base experts' structural learnability guarantees.

Assumption 3 (Smoothing wrapper)**.** *A smoothing wrapper is a possibly randomized mapping υ that takes a finite sequence of decisions $H = \{u^\tau : \tau \in Z\} \subseteq \mathbf{U}$ with $|Z| \geq 1$ and returns a decision $u = \upsilon(H) \in \mathbf{U}$ s.t.*

$$\mathbb{E}_{u\sim\upsilon(H)}\big[L(u) \mid H\big] \;\leq\; \frac{1}{|Z|}\sum_{\tau\in Z} L(u^\tau), \qquad L(u) := \mathbb{E}_{\xi\sim D}[\ell(u,\xi)].$$

Example 1. *Assume $\mathbf{U}$ is convex and $\ell(\cdot,\xi)$ is convex in u for all ξ. A natural choice is the average $u = \frac{1}{|Z|}\sum_{\tau\in Z} u^\tau$, which satisfies Asm. 3 by Jensen's inequality. The randomized form of Asm. 3 also covers non-convex decision spaces: one can uniformly sample u from the history $\{u^\tau : \tau \in Z\}$ [Pacchiano et al.(2020)], in which case the required inequality holds in conditional expectation without requiring convexity.*

Alg. 1 outlines the procedure. Initializing each expert once before the main loop ensures well-defined confidence bounds, costing at most K updates without affecting asymptotic regret. The *Update* step handles both internal learning and bookkeeping (e.g., counters, accumulated loss).

Algorithm 1 M-LCB

Input: experts $\{\mathbf{U}_k\}_{k=1}^K$, per-round budget M, confidence parameter δ, smoothing wrapper υ (see Assumption 3).
Output: (i) sequence of advices $\{u^t\}_{t=1}^T$; (ii) experts trained at each round $\{S_t\}_{t=1}^T$
Initialize each expert with $\texttt{LCB}_k$ and $\texttt{UCB}_k$ from Def. 1.
Warm-start each expert once so that $n_k^t \geq 1$ for all k before the main selection rule is used.
for $t = 1, 2, \ldots, T$ **do**
  for each $k \in [K]$ **do**
    Compute $\texttt{LCB}_k(t,\delta), \texttt{UCB}_k(t,\delta)$
  end for
  $S_t \leftarrow \arg\min_{S\subseteq[K],|S|=M} \sum_{k\in S} \texttt{LCB}_k(t,\delta)$
  $i_t \leftarrow \arg\min_{k\in S_t} \texttt{UCB}_k(t,\delta)$
  $u^t \leftarrow \upsilon(\{u_{i_t}^\tau : \tau \in I_{i_t}(t)\})$ // smoothing over i_t
  Play u^t, suffer $\ell(u^t,\xi^t)$
  for each $k \in S_t$ **do**
    Observe $\ell(u_k^t,\xi^t)$.
    Update expert k.
  end for
end for

4 Theoretical Analysis

This section consolidates the theoretical analysis of the proposed framework. First, we establish the confidence intervals for the online parametric learning models introduced in Section 2.4. We then analyze the performance of the M-LCB algorithm, deriving both the standard and multiple-play cumulative regret bounds. Finally, we present a minimax lower-bound analysis for the considered problem, demonstrating that the M-LCB algorithm achieves order-optimal performance up to logarithmic factors.

4.1 Confidence Interval Construction

In Section 2.4, using the case of online parametric learning as a motivating example, we introduced specific online learning algorithms to act as our base experts. Within this framework, standard online optimization guarantees must be adapted to satisfy the prefix-hindsight regret form required by Assumption 2. Lemma 2 below establishes this connection, demonstrating how deterministic online regret guarantees translate into the required time-uniform confidence bounds.

Lemma 1 (Empirical loss concentration [Shalev-Shwartz and Ben-David(2014), Lemma B.10])**.** *Let $\mathbf{W}$ be a state space and $\mathcal{D}$ a distribution on $\mathbf{E}$. Let $\ell : \mathbf{W}\times\mathbf{E} \to [0,1]$ be a bounded loss function. Fix a predictor $w \in \mathbf{W}$ and define its expected loss $L(w) := \mathbb{E}_{\xi\sim\mathcal{D}}[\ell(w,\xi)]$.*

Then, for any $n > 0$ and $\delta \in (0,1)$, with probability at least $1-\delta$ over the draw of $\{\xi_t\}_{t=1}^n$ i.i.d. from $\mathcal{D}$,

$$\frac{1}{n}\sum_{t=1}^n \ell(w,\xi_t) - L(w) \leq \sqrt{\tfrac{2L(w)\log(1/\delta)}{n}} + \tfrac{2\log(1/\delta)}{3n}.$$

Proof. The claim follows directly from Bernstein's inequality for bounded random variables. □

Lemma 2 (OCO Regret Implies Prefix-Hindsight Regret)**.** *Let $k \in [K]$ be a base expert implementing an online convex optimization (OCO) algorithm. Suppose that for every confidence level $\delta \in (0,1)$, the expert admits a time-uniform high-probability bound $\overline{U}_k(t,\delta)$ on its pseudo-regret $\overline{R}_k(t)$ defined in (3), namely,*

$$\Pr\big(\forall t \geq 1,\ \overline{R}_k(t) \leq \overline{U}_k(t,\delta)\big) \geq 1-\delta.$$

Then the expert satisfies Assumption 2. More precisely, with probability at least $1-\delta$,

$$R_k(t) \leq U_k(t,\delta), \qquad \forall t \geq 1,$$

where

$$U_k(t,\delta) = \overline{U}_k\left(t, \frac{\delta}{2}\right) + t\,G\left(t, \frac{\delta}{4\,t^2}\right),$$

and

$$G(t,\delta_0) = \sqrt{\frac{2\log(1/\delta_0)}{t}} + \frac{2\log(1/\delta_0)}{3t}$$

is the concentration penalty arising from Bernstein's inequality.

Proof. Let $I_k(t)$ be the set of rounds up to time t in which expert k was updated, and let $n = |I_k(t)|$ denote its effective sample size (local time). We reindex these training rounds as $\tau = 1,\ldots,n$, and denote the corresponding decisions and observed losses by u_k^τ and ξ^τ, respectively. For any fixed confidence level $\delta \in (0,1)$, we introduce the concentration penalty function:

$$G(n,\delta') = \sqrt{\frac{2\log(1/\delta')}{n}} + \frac{2\log(1/\delta')}{3n}.$$

Let $\mathcal{E}_1$ be the high-probability event on which the time-uniform OCO guarantee holds for expert k. By the anytime assumption, with probability at least $1-\delta/2$, the following bound holds simultaneously for all $n \geq 1$:

$$\sum_{\tau=1}^{n} \ell(u_k^\tau, \xi^\tau) - \min_{u \in \mathbf{U}_k} \sum_{\tau=1}^{n} \ell(u, \xi^\tau) \leq \overline{U}_k(n, \delta/2). \tag{4}$$

Next, we evaluate the population performance. Let $u_k^\star \in \arg\min_{u \in \mathbf{U}_k} L(u)$ be a fixed population minimizer, so that $L(u_k^\star) = L_k^\star$. Applying the concentration guarantees from Lemma 1 with a localized confidence parameter $\delta_n = \delta/(4n^2)$, we have that for each fixed $n \geq 1$:

$$\frac{1}{n}\sum_{\tau=1}^{n} \ell(u_k^\star, \xi^\tau) \leq L_k^\star + G\big(n, \delta/(4n^2)\big)$$

holds with probability at least $1-\delta/(4n^2)$.

To ensure this property holds uniformly across all possible local time steps, we apply a union bound over $n \geq 1$. Since $\sum_{n=1}^{\infty}(4n^2)^{-1} < 1/2$, the total failure probability is strictly bounded by $\delta/2$. Thus, the uniform concentration event $\mathcal{E}_2$, defined as

$$\sum_{\tau=1}^{n} \ell(u_k^\star, \xi^\tau) \leq nL_k^\star + nG\big(n, \delta/(4n^2)\big) \qquad \forall n \geq 1, \tag{5}$$

holds true with probability at least $1-\delta/2$.

By the intersection bound, the joint event $\mathcal{E}_1 \cap \mathcal{E}_2$ holds with a total probability of at least $1-\delta$. On this joint event, the empirical minimizer cannot perform worse than the specific population optimal decision $u_k^\star$, which implies:

$$\min_{u \in \mathbf{U}_k} \sum_{\tau=1}^{n} \ell(u, \xi^\tau) \leq \sum_{\tau=1}^{n} \ell(u_k^\star, \xi^\tau).$$

Combining this property with the empirical regret control from (4) and the concentration guarantee from (5) leads to the uniform bound:

$$\sum_{\tau=1}^{n} \ell(u_k^\tau, \xi^\tau) \leq nL_k^\star + \overline{U}_k(n, \delta/2) + nG\big(n, \delta/(4n^2)\big) \qquad \forall n \geq 1.$$

Rearranging the terms directly yields the prefix-hindsight regret formulation required by Assumption 2 with the stated choice of U_k, completing the proof. □

4.2 Regret Analysis for M-LCB

In this subsection, we establish the theoretical guarantees for the proposed M-LCB algorithm by deriving upper bounds on its cumulative regret. We begin by defining a uniform high-probability concentration event in Lemma 4, under which all time-uniform confidence intervals hold simultaneously. Building upon this event, Lemma 5 provides a generic pseudo-regret bound expressed in terms of the expert-specific rate functions U_k. Finally, Theorem 1 delivers an explicit rate of convergence under a typical polynomial learning curriculum, where the underlying experts exhibit a regret growth of $O(t^\alpha)$.

Lemma 3 (Azuma's Inequality [Mohri et al.(2018), Theorem D.2])**.** *Let $\{\mathcal{F}_t\}_{t\geq 0}$ be a filtration, and let $\{X_t\}_{t\geq 1}$ be a sequence of real-valued random variables adapted to $\mathcal{F}_t$ such that $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$ and $|X_t| \leq 1$ almost surely for all $t \geq 1$. Then, for any fixed $n \geq 1$ and any confidence level $\delta \in (0,1)$, with probability at least $1-\delta$, the following uniform bound holds:*

$$\left|\frac{1}{n}\sum_{t=1}^{n} X_t\right| \leq \sqrt{\frac{2\log(1/\delta)}{n}}.$$

Lemma 4 (Anytime Confidence Bounds)**.** *For any confidence level $\delta \in (0,1)$, define*

$$\mathcal{E}_\delta := \left\{ \forall k \in [K],\ \forall t \geq 1 : \begin{array}{l} \texttt{LCB}_k(t,\delta) \leq L_k^\star, \\ L_k^\star \leq \texttt{UCB}_k(t,\delta), \\ \mathbb{E}_\upsilon[L(\upsilon(H_k^t)) \mid \mathcal{F}_{t-1}] \leq \texttt{UCB}_k(t,\delta) \end{array} \right\}, \tag{6}$$

where

$$H_k^t := \{u_k^\tau : \tau \in I_k(t)\}$$

denotes the decision history used by the wrapper for expert k up to time t.

Under Assumptions 1–3,

$$\Pr(\mathcal{E}_\delta) \geq 1-\delta.$$

Proof of Lemma 4. Fix an expert $k \in [K]$ and let $n = |I_k(t)|$ denote the number of times it has been updated. Relabel the corresponding update rounds as $\tau = 1, \ldots, n$, with associated decisions u_k^τ and losses $\ell(u_k^\tau, \xi^\tau)$. We first establish the inequalities in (6) for a fixed pair (k,n) and then apply a union bound over all experts and update counts.

By Assumption 2, with probability at least $1-\delta_{\text{arm}}$, the anytime regret guarantee holds simultaneously for all $n \geq 1$, namely

$$\frac{1}{n}\sum_{\tau=1}^{n} \ell(u_k^\tau, \xi^\tau) - \frac{U_k(n, \delta_{\text{arm}})}{n} \leq L_k^\star. \tag{7}$$

Evaluating (7) at $n = n_k^t$ and recalling the definition of $\texttt{LCB}_k(t,\delta)$ immediately yields $\texttt{LCB}_k(t,\delta) \leq L_k^\star$.

To obtain the upper confidence bound, consider the filtration $\mathcal{H}_k^{n-1} = \{(u_k^\tau, \xi^\tau) : \tau \in [n-1]\}$. Since u_k^n is measurable with respect to $\mathcal{H}_k^{n-1}$ and ξ^n is independent of the past,

$$\mathbb{E}\big[\ell(u_k^n, \xi^n) \,\big|\, \mathcal{H}_k^{n-1}\big] = L(u_k^n).$$

Consequently, $\{\ell(u_k^\tau, \xi^\tau) - L(u_k^\tau)\}_{\tau=1}^n$ forms a bounded martingale-difference sequence. Applying Lemma 3 with confidence level δ_n gives, with probability at least $1-\delta_n$,

$$\left|\frac{1}{n}\sum_{\tau=1}^{n} \ell(u_k^\tau, \xi^\tau) - \frac{1}{n}\sum_{\tau=1}^{n} L(u_k^\tau)\right| \leq H(n, \delta_n). \tag{8}$$

Let

$$L_k(n) := \frac{1}{n}\sum_{\tau=1}^{n} \ell(u_k^\tau, \xi^\tau).$$

Combining (8) with the fact that $L_k^\star = \min_{u \in \mathbf{U}_k} L(u) \leq L(u_k^\tau)$ for every τ yields

$$L_k^\star \leq \frac{1}{n}\sum_{\tau=1}^{n} L(u_k^\tau) \leq L_k(n) + H(n, \delta_n).$$

Evaluating this inequality at $n = n_k^t$ gives $L_k^\star \le \texttt{UCB}_k(t,\delta)$.

Next, Assumption 3 implies that the wrapper does not increase the average expected loss. Applied to the history $\{u_k^\tau\}_{\tau=1}^n$, it yields

$$\mathbb{E}_\upsilon\left[L(\upsilon(H_k^t)) \,\middle|\, H_k^t\right] \le \frac{1}{n}\sum_{\tau=1}^{n} L(u_k^\tau).$$

Combining this with (8) gives

$$\mathbb{E}_\upsilon\left[L(\upsilon(H_k^t)) \,\middle|\, H_k^t\right] \le L_k(n) + H(n,\delta_n) = \texttt{UCB}_k(t,\delta),$$

which establishes the third inequality in (6).

Finally, we union bound over all experts and update counts. The failure probability associated with the anytime guarantees (7) is at most

$$K\delta_{\mathrm{arm}} = \frac{\delta}{2}.$$

For the martingale concentration events,

$$\sum_{k=1}^{K}\sum_{n=1}^{\infty}\delta_n = \sum_{k=1}^{K}\sum_{n=1}^{\infty}\frac{\delta}{4Kn^2} = \frac{\delta}{4}\sum_{n=1}^{\infty}\frac{1}{n^2} < \frac{\delta}{2}.$$

Therefore, by another union bound, all inequalities in (6) hold simultaneously for every expert and every time step with probability at least $1-\delta$.

□

Remark 2. *In the algebraic verification of Lemma 4, the concentration bounds for bounded losses can be sharpened tightly by introducing a truncated upper confidence bound $Z_k(t,\delta) := \min\{1, \texttt{UCB}_k(t,\delta)\}$. Consequently, the variance-dependent concentration penalty function $G_k(t,\delta)$ can be modified as:*

$$G_k(t,\delta) := \sqrt{\frac{2Z_k(t,\delta)\log(1/\delta)}{3t}} + \frac{2\log(1/\delta)}{t}.$$

4.2.1 Regret Guarantees

Lemma 5 (Regret Bound)**.** *Let the pseudo-regret be*

$$\widetilde{\mathrm{Reg}}(T) := \mathbb{E}_\upsilon\left[\sum_{t=1}^{T} L(u^t)\right] - TL^\star,$$

where the expectation is taken over the randomization of the smoothing wrapper. Fix $\delta \in (0,1)$ and suppose that Assumptions 1– 3 hold. Then, on the event $\mathcal{E}_\delta$ (6), for all $T \ge 1$,

$$\widetilde{\mathrm{Reg}}(T) \le \mathbb{E}_\upsilon[\Delta(T)],$$

where

$$\Delta(T) := \sum_{\tau\in I_{k^\star}(T)}\Big(\texttt{UCB}_{k^\star}(\tau,\delta) - \texttt{LCB}_{k^\star}(\tau,\delta)\Big) + \frac{1}{M}\sum_{k=1}^{K}\sum_{\tau\in I_k(T)}\Big(\texttt{UCB}_k(\tau,\delta) - \texttt{LCB}_k(\tau,\delta)\Big), \tag{9}$$

and

$$k^\star \in \arg\min_{k\in[K]} L_k^\star.$$

Moreover, with probability at least $1-2\delta$, uniformly over all $T \ge 1$,

$$\mathrm{Reg}(T) \le \mathbb{E}_\upsilon[\Delta(T)] + O\Big(\sqrt{T\log(1/\delta)}\Big).$$

Proof of Lemma 5. **Pseudo-regret bound**. S_t and i_t denote, respectively, the training subset and the prediction arm selected at round t by Alg. 1. Let $k^\star \in \arg\min_{k\in[K]} L^\star_k$ be the index of the best arm in terms of expected loss.

Algorithm uses the confidence bounds from Prop. 4, so $\mathcal{E}_\delta$ (6) holds with probability at least $1-\delta$. In the sequel, we condition on $\mathcal{E}_\delta$ and prove the regret bounds under this event. Under event $\mathcal{E}_\delta$, the safe advice at time t is $u^t = v(\{u^\tau_{i_t} : \tau \in I_{i_t}(t)\})$. By Asm. 3 and the definition of $\mathcal{E}_\delta$, its conditional expected loss is bounded by $\mathtt{UCB}_{i_t}(t,\delta)$. Therefore,

$$\widetilde{\mathrm{Reg}}(T) = \mathbb{E}_v\left[\sum_{t=1}^{T}\big[L(u^t) - L^\star\big]\right] \le \mathbb{E}_v\left[\sum_{t=1}^{T}\big[\mathtt{UCB}_{i_t}(t,\delta) - L^\star\big]\right],$$

where the outer expectation is omitted because all subsequent inequalities hold pointwise for each realized sequence of wrapper choices. Splitting the sum depending on whether $k^\star$ is trained at t gives

$$\overline{\mathrm{Reg}}(T) \le \underbrace{\sum_{t:\, k^\star\in S_t}\big[\mathtt{UCB}_{i_t}(t,\delta) - L^\star\big]}_{\text{A}} + \underbrace{\sum_{t:\, k^\star\notin S_t}\big[\mathtt{UCB}_{i_t}(t,\delta) - L^\star\big]}_{\text{B}}. \tag{10}$$

Term A. Since $i_t = \arg\min_{k\in S_t}\mathtt{UCB}_k(t,\delta)$, for rounds with $k^\star \in S_t$, $\mathtt{UCB}_{i_t}(t,\delta) \le \mathtt{UCB}_{k^\star}(t,\delta)$.

Under $\mathcal{E}_\delta$, $L^\star \ge \mathtt{LCB}_{k^\star}(t,\delta)$. Therefore,

$$\text{A} \le \sum_{t:\, k^\star\in S_t}\Big[\mathtt{UCB}_{k^\star}(t,\delta) - \mathtt{LCB}_{k^\star}(t,\delta)\Big] = \sum_{\tau\in I_{k^\star}(T)}\Big[\mathtt{UCB}_{k^\star}(\tau,\delta) - \mathtt{LCB}_{k^\star}(\tau,\delta)\Big] \tag{11}$$

The last equality is from fact that k^* is updated in such and only such terms of sum.

Term B. By construction of i_t, $\mathtt{UCB}_{i_t}(t,\delta) \le \frac{1}{M}\sum_{k\in S_t}\mathtt{UCB}_k(t,\delta)$. Substituting this into term B in (10) and applying add–subtract trick with $\frac{1}{M}\sum_{k\in S_t}\mathtt{LCB}_k(t,\delta)$ for each t, we obtain:

$$B \le \underbrace{\sum_{t:k^\star\notin S_t}\left[\frac{1}{M}\sum_{k\in S_t}\mathtt{LCB}_k(t,\delta) - L^\star\right]}_{\text{C}} + \underbrace{\sum_{t:k^\star\notin S_t}\frac{1}{M}\sum_{k\in S_t}\Big[\mathtt{UCB}_k(t,\delta) - \mathtt{LCB}_k(t,\delta)\Big]}_{\text{D}}.$$

Term C. Since S_t consists of the M smallest LCBs and $k^\star \notin S_t$, every $k\in S_t$ has $\mathtt{LCB}_k(t,\delta) \le \mathtt{LCB}_{k^\star}(t,\delta) \le L^\star$; hence $\frac{1}{M}\sum_{k\in S_t}\mathtt{LCB}_k(t,\delta) \le L^\star$. Thus that term is non-positive. Therefore,

$$\text{B} \le \sum_{t:k^\star\notin S_t}\frac{1}{M}\sum_{k\in S_t}\Big[\mathtt{UCB}_k(t,\delta) - \mathtt{LCB}_k(t,\delta)\Big] \le \frac{1}{M}\sum_{k=1}^{K}\sum_{\tau\in I_k(T)}\Big[\mathtt{UCB}_k(\tau,\delta) - \mathtt{LCB}_k(\tau,\delta)\Big]. \tag{12}$$

Combining (11) and (12) with (10) proves the best-arm bound stated in the proposition.

Regret bound. The proof follows by decomposing regret into a stochastic term and the pseudo-regret:

$$\mathrm{Reg}(T) = \sum_{t=1}^{T}\Big(\ell(u^t,\xi^t) - \mathbb{E}_v L(u^t)\Big) + \mathbb{E}_v\sum_{t=1}^{T}\Big(L(u^t) - L^\star\Big).$$

Since the data are independent of u^t conditional on the past, the first term is a sum of martingale difference sequence $Z_t = \ell(u^t,\xi^t) - \mathbb{E}_v[L(u^t) \mid \mathcal{F}_{t-1}]$ and lies in $[-1,1]$, so it bounded by a concentration inequality (e.g., Lemma 3). The second term is the pseudo-regret bounded above. $\square$

Theorem 1 (Convergence Rates)**.** *Let $\alpha,\delta\in(0,1)$ and suppose that every expert $k\in[K]$ satisfies Assumption 2 with*

$$U_k(t,\delta) = O\big(t^\alpha c(\delta)\big).$$

Then, with probability at least $1-\delta$, the regret of Alg. 1 satisfies, uniformly over all $T\ge 1$

$$\mathrm{Reg}(T) = O\left(\sqrt{\frac{KT}{M}\log\frac{KT}{\delta}} + \Big(\frac{K}{M}\Big)^{1-\alpha}T^\alpha c(\delta)\right).$$

Proof of Theorem 1. By Lemma 5, it suffices to bound the quantity $\Delta(T)$ appearing in (9). Define

$$\Delta_k(n,\delta) := H(n,\delta_n) + \frac{U_k(n,\delta_{\mathrm{arm}})}{n}.$$

From Definition 1, $\texttt{UCB}_k(t,\delta) - \texttt{LCB}_k(t,\delta) = \Delta_k(n_k^t,\delta)$, and therefore

$$\Delta(T) = \sum_{\tau=1}^{n_{k^\star}^T} \Delta_{k^\star}(\tau,\delta) + \frac{1}{M}\sum_{k=1}^{K}\sum_{\tau=1}^{n_k^T} \Delta_k(\tau,\delta).$$

Using the definition $\delta_n = \delta/(4Kn^2)$ and the fact that $n \le T$, we obtain $H(n,\delta_n) = O\left(\sqrt{\frac{\log(KT/\delta)}{n}}\right)$. Moreover, Assumption 2 together with $U_k(t,\delta) = O\big(t^\alpha c(\delta)\big)$ yields $\frac{U_k(n,\delta_{\mathrm{arm}})}{n} = O\big(n^{\alpha-1}c(\delta)\big)$. Hence,

$$\Delta_k(n,\delta) = O\left(\sqrt{\frac{\log(KT/\delta)}{n}} + n^{\alpha-1}c(\delta)\right).$$

Substituting this estimate into the expression for $\Delta(T)$, we need to control sums of the form $\sum_{\tau=1}^{n}\tau^{-1/2}$ and $\sum_{\tau=1}^{n}\tau^{\alpha-1}$. Using the standard bounds $\sum_{\tau=1}^{n}\tau^{-1/2} = O(\sqrt{n})$ and $\sum_{\tau=1}^{n}\tau^{\alpha-1} = O(n^\alpha)$, we obtain

$$\sum_{\tau=1}^{n_{k^\star}^T} \Delta_{k^\star}(\tau,\delta) = O\left(\sqrt{T\log\frac{KT}{\delta}} + T^\alpha c(\delta)\right).$$

For the second term, using Cauchy–Schwarz together with $\sum_{k=1}^{K} n_k^T \le MT$, we obtain

$$\frac{1}{M}\sum_{k=1}^{K}\sqrt{n_k^T} \le \frac{\sqrt{K\sum_{k=1}^{K} n_k^T}}{M} = O\left(\sqrt{\frac{KT}{M}}\right).$$

Similarly, since $x \mapsto x^\alpha$ is concave for $\alpha \in (0,1)$, Jensen's inequality gives

$$\frac{1}{M}\sum_{k=1}^{K}(n_k^T)^\alpha \le \frac{1}{M}K^{1-\alpha}\left(\sum_{k=1}^{K} n_k^T\right)^\alpha \le O\left(\left(\frac{K}{M}\right)^{1-\alpha} T^\alpha\right).$$

Combining the above estimates yields

$$\Delta(T) = O\left(\sqrt{\frac{KT}{M}\log\frac{KT}{\delta}} + \left(\frac{K}{M}\right)^{1-\alpha} T^\alpha c(\delta)\right).$$

Applying Lemma 5, we obtain the same bound for the pseudo-regret. Finally, the additional martingale term in Lemma 5 contributes at most $O\left(\sqrt{T\log(1/\delta)}\right)$, which is dominated by $O\left(\sqrt{\frac{KT}{M}\log\frac{KT}{\delta}}\right)$ since $K \ge M$. The stated regret bound follows. $\square$

4.2.2 Multiple-Play Regret Guarantees

The M-LCB framework naturally extends to the multiple-play bandit setting [Agrawal et al.(1990), Uchiya et al.(2010)]. At each round, the algorithm constructs the subset S_t and deploys all experts rather than selecting a single representative expert via the UCB rule. The incurred loss is $\frac{1}{M}\sum_{k\in S_t} L_k^\star$, and all experts in S_t are updated using the observed feedback. Performance is measured relative to the best subset of M experts in hindsight. This setting is closely related to combinatorial and multiple-play bandits, where UCB-based methods are known to achieve $\widetilde{O}(\sqrt{KT/M})$ regret under fixed stochastic rewards [Kveton et al.(2015)].

Lemma 6 (Top-M experts regret)**.** *Fix $\delta \in (0,1)$ and suppose the assumptions of Lemma 5 hold. Let*

$$\overline{L}^\star := \min_{S\subseteq[K],\,|S|=M} \frac{1}{M}\sum_{k\in S} L_k^\star$$

denote the average optimal loss of the best subset of M experts. Define the Top-M regret as

$$\mathrm{Reg}_M(T) := \sum_{t=1}^{T} \left[\frac{1}{M} \sum_{k \in S_t} L_k^\star - \overline{L}^\star \right].$$

Then, with probability at least $1-\delta$,

$$\mathrm{Reg}_M(T) \le \frac{1}{M} \sum_{k=1}^{K} \sum_{\tau \in I_k(T)} \big[\mathit{UCB}_k(\tau,\delta) - \mathit{LCB}_k(\tau,\delta) \big].$$

Proof of Lemma 6. Let

$$S^\star \in \arg \min_{S \subseteq [K],\, |S|=M} \frac{1}{M} \sum_{k \in S} L_k^\star.$$

On the event $\mathcal{E}_\delta$, Lemma 4 ensures that

$$\mathtt{LCB}_k(t,\delta) \le L_k^\star \le \mathtt{UCB}_k(t,\delta) \qquad \forall k \in [K],\ \forall t \ge 1.$$

Since S_t is chosen as the subset minimizing the sum of lower confidence bounds,

$$\frac{1}{M} \sum_{k \in S_t} \mathtt{LCB}_k(t,\delta) \le \frac{1}{M} \sum_{k \in S^\star} \mathtt{LCB}_k(t,\delta) \le \frac{1}{M} \sum_{k \in S^\star} L_k^\star = \overline{L}^\star.$$

Therefore,

$$\begin{aligned} \frac{1}{M} \sum_{k \in S_t} L_k^\star - \overline{L}^\star &\le \frac{1}{M} \sum_{k \in S_t} \big(L_k^\star - \mathtt{LCB}_k(t,\delta) \big) \\ &\le \frac{1}{M} \sum_{k \in S_t} \big(\mathtt{UCB}_k(t,\delta) - \mathtt{LCB}_k(t,\delta) \big). \end{aligned}$$

Summing over $t = 1, \ldots, T$ and regrouping terms according to the update indices of each expert yields

$$\mathrm{Reg}_M(T) \le \frac{1}{M} \sum_{k=1}^{K} \sum_{\tau \in I_k(T)} \big[\mathtt{UCB}_k(\tau,\delta) - \mathtt{LCB}_k(\tau,\delta) \big],$$

which proves the claim.

□

Theorem 2 (Convergence rate for Top-M regret)**.** *Let $\alpha, \delta \in (0,1)$ and suppose that every expert $k \in [K]$ satisfies Assumption 2 with*

$$U_k(t,\delta) = O\big(t^\alpha c(\delta)\big).$$

Then, with probability at least $1-\delta$, the Top-M regret satisfies uniformly over all $T \ge 1$,

$$\mathrm{Reg}_M(T) = O\left(\sqrt{\frac{KT}{M} \log \frac{KT}{\delta}} + \left(\frac{K}{M}\right)^{1-\alpha} T^\alpha c(\delta) \right).$$

Proof of Theorem 2. By Lemma 6, the regret is controlled by the same confidence-width sum that appears in Lemma 5, except that the contribution of the optimal expert is absent. Consequently, the proof of Theorem 1 applies without modification.

Indeed, each confidence width satisfies

$$\mathtt{UCB}_k(t,\delta) - \mathtt{LCB}_k(t,\delta) = O\left(\sqrt{\frac{\log(KT/\delta)}{n_k^t}} + (n_k^t)^{\alpha-1} c(\delta) \right).$$

Summing the concentration term via Cauchy–Schwarz and the learning term via the concavity bound

$$\sum_{k=1}^{K} (n_k^T)^\alpha \le K^{1-\alpha} (MT)^\alpha$$

yields the stated rate.

□

Remark 3 (On constants)**.** *Suppose the individual expert guarantees take the form*

$$U_k(t,\delta) = O\big(\beta_k\, t^{\alpha} c(\delta)\big),$$

where the constants β_k may depend on problem-specific quantities such as dimension, smoothness, or Lipschitz parameters. Repeating the proof while tracking these constants yields the refined bound

$$O\left(\sqrt{\frac{KT}{M}\log\frac{KT}{\delta}} + T^{\alpha}c(\delta)\left[\beta_{k^\star} + \left(\frac{1}{M}\sum_{k=1}^{K}\beta_k^{\frac{1}{1-\alpha}}\right)^{1-\alpha}\right]\right),$$

which more precisely captures the dependence on the complexity of the underlying experts.

4.3 Minimax Lower Bounds

We now show that the regret guarantees obtained in Theorems 1 and 2 are essentially unimprovable. The lower bound combines two sources of difficulty. The first is the exploration problem of identifying the best expert among K candidates when only M experts can be updated at each round. The second is the intrinsic statistical complexity of the underlying learning tasks solved by the experts.

To formalize the latter component, we introduce a family of α-hard learning problems.

Definition 2 (α-hard class)**.** *A class of learning problems $\mathcal{F}_\alpha$ is called α-hard if there exists a constant $c > 0$ such that, for every algorithm and every horizon n, one can find $f \in \mathcal{F}_\alpha$ satisfying*

$$R^{\mathrm{in}}(n) \geq c\, n^{\alpha},$$

where $R^{\mathrm{in}}(n)$ denotes the internal regret of the learner on problem f.

The construction used in the proof is based on heavy-tailed stochastic bandits introduced by [Bubeck et al.(2013)]. These problems satisfy $\alpha = \frac{1}{1+\beta}$, where β characterizes the heaviness of the tails.

Theorem 3 (Minimax lower bound)**.** *Let K be the number of experts, M the per-round update budget, and T the horizon. For every $\alpha \in [0,1]$, there exists an α-hard class $\mathcal{F}_\alpha$ such that, for any collection of learning algorithms $\{\mathscr{A}_k\}_{k=1}^{K}$ and any meta-procedure,*

$$\sup_{f_k \in \mathcal{F}_\alpha} \mathbb{E}\,\mathrm{Reg}(T) \geq c_1\sqrt{\frac{KT}{M}} + c_2 T^{\alpha}\left(\frac{K}{M}\right)^{1\alpha},$$

provided $\sqrt{\frac{K\log K}{MT}}$ is sufficiently small. The constants $c_1, c_2 > 0$ are universal.

Proof sketch. The proof is based on a family of perturbed games together with a symmetric null game. The exploration component is obtained using a standard information-theoretic argument. Pinsker's inequality relates the probability of identifying the optimal expert to the KL divergence between the null and perturbed games, yielding the term $\Omega\Big(\sqrt{\frac{KT}{M}}\Big)$.

To obtain the second term, we construct each expert as an α-hard learning problem. A concentration argument shows that, in the null game, every expert receives approximately MT/K updates. The corresponding internal regret therefore contributes $\Omega\Big(T^{\alpha}\left(\frac{K}{M}\right)^{1-\alpha}\Big)$. Combining the two components proves the claim. Full details are provided in Appendix 4.3. □

The lower bound matches the upper bounds of Theorems 1 and 2 up to logarithmic factors. Consequently, the dependence on the horizon T, the number of experts K, the update budget M, and the hardness parameter α cannot be improved in general. In particular, the term $\sqrt{KT/M}$ is the unavoidable price of expert identification, whereas $T^{\alpha}(K/M)^{1-\alpha}$ captures the intrinsic complexity of learning within the experts.

5 Discussion and Examples

Theorems 1 and 2 reveal a simple decomposition of the regret of M-LCB. The first term, $\tilde{O}\Big(\sqrt{\frac{KT}{M}}\Big)$, arises from the need to identify promising experts under a limited training budget. Its dependence on K, M, and T

Table 2: Examples of inner-arm convergence rates $U_k(n,\delta)$ and resulting global regret (up to logarithmic factors and an additive term $O(\sqrt{(K/M)T})$). For each expert k, the inner algorithm satisfies $U_k(t,\delta) = O(\beta_k\, t^\alpha c(\delta))$, and the corresponding global regret scales as $O\big(T^\alpha c(\delta)\, \|\boldsymbol{\beta}\|_{M,1-\alpha}\big)$, where $\|\boldsymbol{\beta}\|_{M,\gamma} = \Big(\frac{1}{M}\sum_{k=1}^K \beta_k^{\frac{1}{\gamma}}\Big)^\gamma$. Parameter conventions: K — # experts; M — per-round training budget; T — horizon; N_k — # base arms/actions; d_k — feature dimension; ε — heavy-tail moment exponent ($\mathbb{E}|X|^{1+\varepsilon} \le \sigma^{1+\varepsilon}$); L, R, C, G — Lipschitz, diameter, range, and gradient constants.

Inner algorithm / problem	**Inner rate** $U_k(n,\delta)$	**Global regret (up to logs)**
OGD / OMD (convex Lipschitz)	$O(G_k R_k \sqrt{n})$, $\alpha = \frac{1}{2}$,	$O\big(T^{1/2}\, \|GD\|_{M,1/2}\big)$
Bandit Convex Optimization (bounded f_t) [Flaxman et al.(2004)]	$O(C_k d_k n^{5/6})$, $\alpha = \frac{5}{6}$	$O\big(T^{5/6}\, \|Cd\|_{M,1/6}\big)$
Bandit Convex Optimization (L–Lipschitz f_t) [Flaxman et al.(2004)]	$O(\sqrt{C_k L_k R_k} d_k n^{3/4})$, $\alpha = \frac{3}{4}$	$O\big(T^{3/4}\, \|\sqrt{CLR}d\|_{M,1/4}\big)$
Heavy–tailed stochastic bandits [Bubeck et al.(2013)]	$\tilde{O}(n^\alpha N_k^{1-\alpha})$, $\alpha = \frac{1}{1+\varepsilon}$	$O(T^\alpha\, [\frac{1}{M}\sum_{k=1} N_k]^{1-\alpha})$
Heavy–tailed stochastic bandits (Symmetric noise) [Dorn et al.(2025a)]	$O(\sqrt{N_k n} \log n^{\frac{3}{2}})$, $\alpha = \frac{1}{2}$	$O(T^{1/2}\, [\frac{1}{M}\sum_{k=1} N_k]^{\frac{1}{2}})$
Heavy–tailed linear bandits [Tajdini et al.(2025)]	$O(d_k^{\frac{3}{2}-\alpha} n^\alpha)$, $\alpha = \frac{1}{1+\varepsilon}$	$O\big(T^\alpha\, \|d^{\frac{3}{2}-\alpha}\|_{M,1-\alpha}\big)$
Hedge / Exponential Weights (over N_k experts)	$O(\sqrt{n \log N_k})$, $\alpha = \frac{1}{2}$,	$O\big(T^{1/2}\, \|\sqrt{\log N}\|_{M,1/2}\big)$

coincides, up to logarithmic factors, with the rates known for stochastic multiple-play bandits. The second term, $\tilde{O}\Big(\big(\frac{K}{M}\big)^{1-\alpha} T^\alpha c(\delta)\Big)$, reflects the learning dynamics of the underlying experts and is inherited directly from their individual convergence guarantees.

The parameter α characterizes the difficulty of the inner learning problem. When α is small, experts learn rapidly and the exploration term becomes the dominant component of the regret. In this regime, the performance of M-LCB approaches that of classical multiple-play bandit algorithms. As α increases, the overall regret becomes increasingly influenced by the convergence properties of the experts themselves.

A comparison with existing approaches is presented in Table 1. Classical model-selection methods such as CORRAL [Pacchiano et al.(2020)] and Dynamic Balancing [Cutkosky et al.(2021)] are designed for learnable experts and achieve strong regret guarantees in the single-update setting. However, they allocate resources to only one expert per round and do not address subset-level performance. On the other hand, multiple-play bandit algorithms [Seldin et al.(2014), Yun et al.(2018), Kveton et al.(2015)] explicitly exploit a budget of M actions per round, but assume that the arms are fixed and do not improve through training. The M-LCB framework combines these two perspectives. It supports arbitrary learnable experts while allowing up to M experts to be updated at every iteration. At the same time, it provides guarantees for both the single-action regret considered in Section 4.2.1 and the Top-M regret introduced in Section 4.2.2. The resulting rates preserve the optimal $\tilde{O}(\sqrt{KT/M})$ exploration dependence of multiple-play methods while incorporating the statistical complexity of the underlying learning procedures.

Table 2 illustrates the implications of the theory for several representative expert classes. A particularly important case is online convex optimization, where many standard algorithms satisfy $U_k(T) = O(\sqrt{T})$, corresponding to $\alpha = \frac{1}{2}$. In this setting, the exploration and learning contributions have the same order and jointly determine the final regret rate. Stronger assumptions, such as strong convexity or realizability, lead to faster convergence, whereas heavy-tailed or otherwise difficult stochastic environments result in slower rates.

Another advantage of the framework is its ability to handle heterogeneous collections of experts. In practical applications, experts may differ substantially in their action spaces, model classes, dimensions, or smoothness assumptions. Such differences are captured by the constants β_k appearing in Remark 3. Rather than depending on the complexity of the most difficult expert, the resulting regret bound depends on an aggregate measure of expert complexity, allowing M-LCB to adapt naturally to diverse families of learning algorithms.

6 Experiments

We evaluate the empirical performance of M-LCB on heterogeneous model-selection problems with adaptive experts. The experiments are designed to assess how effectively the proposed method utilizes a limited update budget, how quickly it identifies high-performing experts, and how its behavior compares with existing approaches for learnable experts and budgeted expert advice. We consider two complementary benchmarks. The first is a heterogeneous stochastic bandit model-selection problem that highlights the interaction between expert quality and learning speed. The second is a generalized linear model (GLM) selection task in which the candidate models differ through their underlying link functions.

6.1 Model Selection among Stochastic Bandits

We first consider a heterogeneous multi-armed bandit benchmark designed to create a "suboptimal trap" for meta-learning algorithms. The benchmark contains $K = 10$ experts, each controlling a two-armed Bernoulli bandit.

For expert $k \in \{0, \dots, K-1\}$, the expected rewards are

$$\mu_{k,1} = \frac{2}{10}\left(1 - \frac{k}{K}\right), \qquad \mu_{k,2} = \frac{1}{10}\left(2 - \frac{k}{K}\right).$$

As k increases, the maximum achievable reward decreases, while the gap between the two arms increases. Consequently, lower-quality experts are easier to optimize and often appear attractive during the early stages of learning.

To introduce heterogeneous learning dynamics, the first $K/2$ experts use EpsilonGreedy as the internal learner, whereas the remaining experts use UCB1. This setup intentionally creates a situation where rapidly converging but ultimately suboptimal experts may initially outperform the optimal expert.

We compare M-LCB against LimitedAdvice and Smooth Corral, a smoothed variant of Corral [Agarwal et al.(2017)]. For M-LCB and LimitedAdvice, we consider update budgets $M \in \{1, 2, 3, 4\}$. Each experiment is run for $T = 2.5 \times 10^5$ rounds and averaged over 100 independent repetitions.

Results. Figure 1 summarizes the results. Panel (a) shows cumulative regret. All methods achieve sublinear growth, while both M-LCB and LimitedAdvice benefit substantially from larger update budgets. Panel (b) reports the final expert-selection distribution. As M increases, M-LCB concentrates more rapidly on the optimal expert, indicating that additional optimization budget accelerates identification of the best learner. Panel (c) illustrates how the computational budget is allocated. Unlike methods that distribute updates more uniformly,M-LCB focuses training on promising experts. This allows strong experts to overcome their initial exploration phase more quickly and prevents the meta-learner from being trapped by suboptimal experts with favorable short-term performance.

6.2 Model Selection among Generalized Linear Models

We next evaluate M-LCB on a synthetic model-selection task involving generalized linear models (GLMs).

We compare against two baselines: $\texttt{ED}^2\texttt{RB}$ [Dann et al.(2024)], which provides guarantees for learnable experts, and LimitedAdvice [Seldin et al.(2014)], which supports multiple updates per round. We consider update budgets $M \in \{1, 2, 3\}$, average over 30 independent runs, and display ± 0.5 standard deviations as shaded regions.

The benchmark contains $K = 10$ candidate models. Each arm corresponds to a generalized linear model with its own nonlinearity $f_k : \mathbb{R} \to \mathbb{R}$.

At round t, a feature vector $x_t \in \mathbb{S}^{d-1}$ is sampled uniformly from the unit sphere, and the target value is generated according to the optimal model $k^\star = 9$: $r_t = f_{k^\star}(w^\top x_t)$.

Figure 2 visualizes the nonlinearities. The final three models are intentionally chosen to be very similar in high-density regions of the feature space, making identification of the optimal model nontrivial.

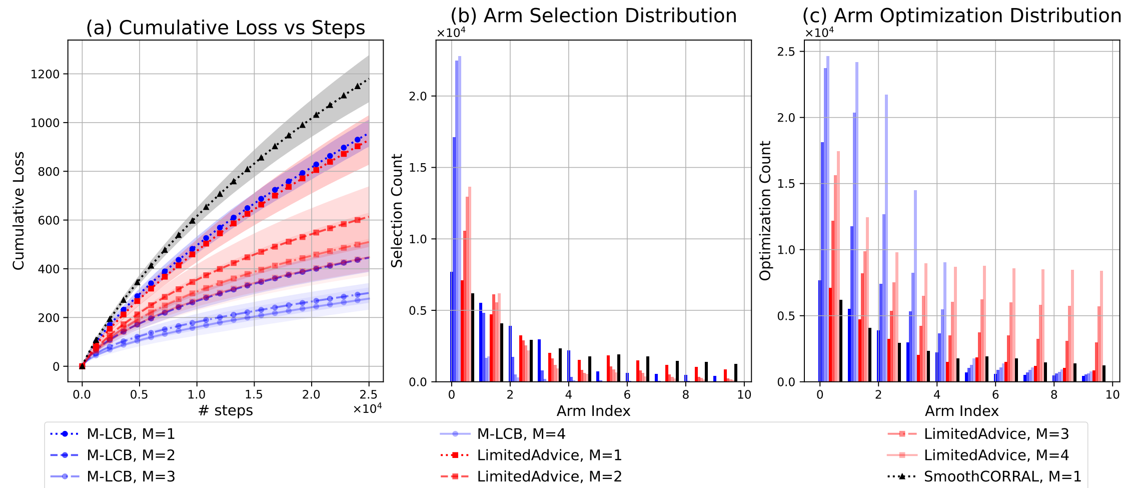

Figure 1: Performance comparison on the heterogeneous bandit model-selection problem. (a) Cumulative regret. (b) Final expert-selection distribution. (c) Allocation of computational budget across experts.

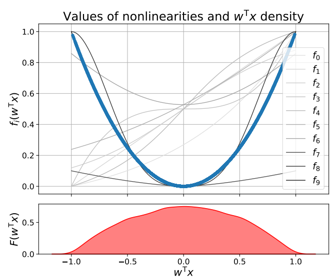

Figure 2: Link functions associated with the candidate models (top) and the density of generated samples (bottom). The last three models are highly similar in the region where most observations occur, making model selection difficult.

Results. Figure 3 reports the results. Panel (a) shows cumulative regret and demonstrates that M-LCB remains competitive with both baselines. Panel (b) presents the final model-selection frequencies. The algorithm consistently identifies the optimal model ($k = 9$), despite the similarity of the strongest candidates. Panel (c) illustrates how optimization effort is distributed. M-LCB concentrates updates on the most promising models, whereas LimitedAdvice spreads the budget more uniformly. This targeted allocation leads to more efficient use of the available training budget.

Hyperparameters

For $\texttt{ED}^2\texttt{RB}$, the exploration parameter was tuned, with $c = 0.1$ giving the best results. For M-LCB, concentration terms was once tuned in validation environment. Parameters for LimitedAdvice were set according to its theoretical analysis [Seldin et al.(2014)]. Analysis shows that regret is not highly sensitive to this parameters.

7 Limitations and Future Work

Our analysis assumes that the confidence scaling function $c(\delta)$ is known. In the bound of Theorem 1, this function appears only through the expert-learning term $(K/M)^{1-\alpha}T^{\alpha}c(\delta)$. Consequently, when $\alpha < 1/2$ this contribution is asymptotically dominated by the exploration term, whereas for larger α the dependence on $c(\delta)$ is linear. In many standard experts, including OGD, UCB, and Hedge, $c(\delta)$ is polylogarithmic. Practical tuning of the concentration bonus can be mitigated by replacing the Hoeffding-style step with self-normalized bounds when the base experts admit them, but the dependence on the base experts' confidence calibration is inherent to UCB-type methods.

The case of unknown $c(\delta)$ is considered by Pacchiano et al. [Pacchiano et al.(2020)], where the resulting regret bounds exhibit a weaker dependence on $c(\delta)$, while Dann et al. [Dann et al.(2024)] estimate it online, yielding adaptive but less interpretable guarantees. Adapting M-LCB to unknown or misspecified expert-rate functions is therefore an important direction for future work.

Another important direction concerns algorithms with additional observations per round, such as limited-advice and multi-play bandits [Seldin et al.(2014), Yun et al.(2018), Kveton et al.(2015)]. Extending their analysis to the setting of *learnable experts* would unify these frameworks.

Another open question is whether meaningful gap-dependent logarithmic guarantees can be obtained. In classical stochastic bandits, the suboptimality gap is fixed, enabling $O(\log T/\Delta)$-type bounds. In our setting the instantaneous regret $L(u_k^t) - L^\star$ changes as experts learn, so the relevant gap is time-dependent and a direct analogue is not immediate.

Finally, a promising extension is the *contextual* regime, where experts specialize based on observed features or domains, connecting our framework to contextual bandits [Foster et al.(2019)].

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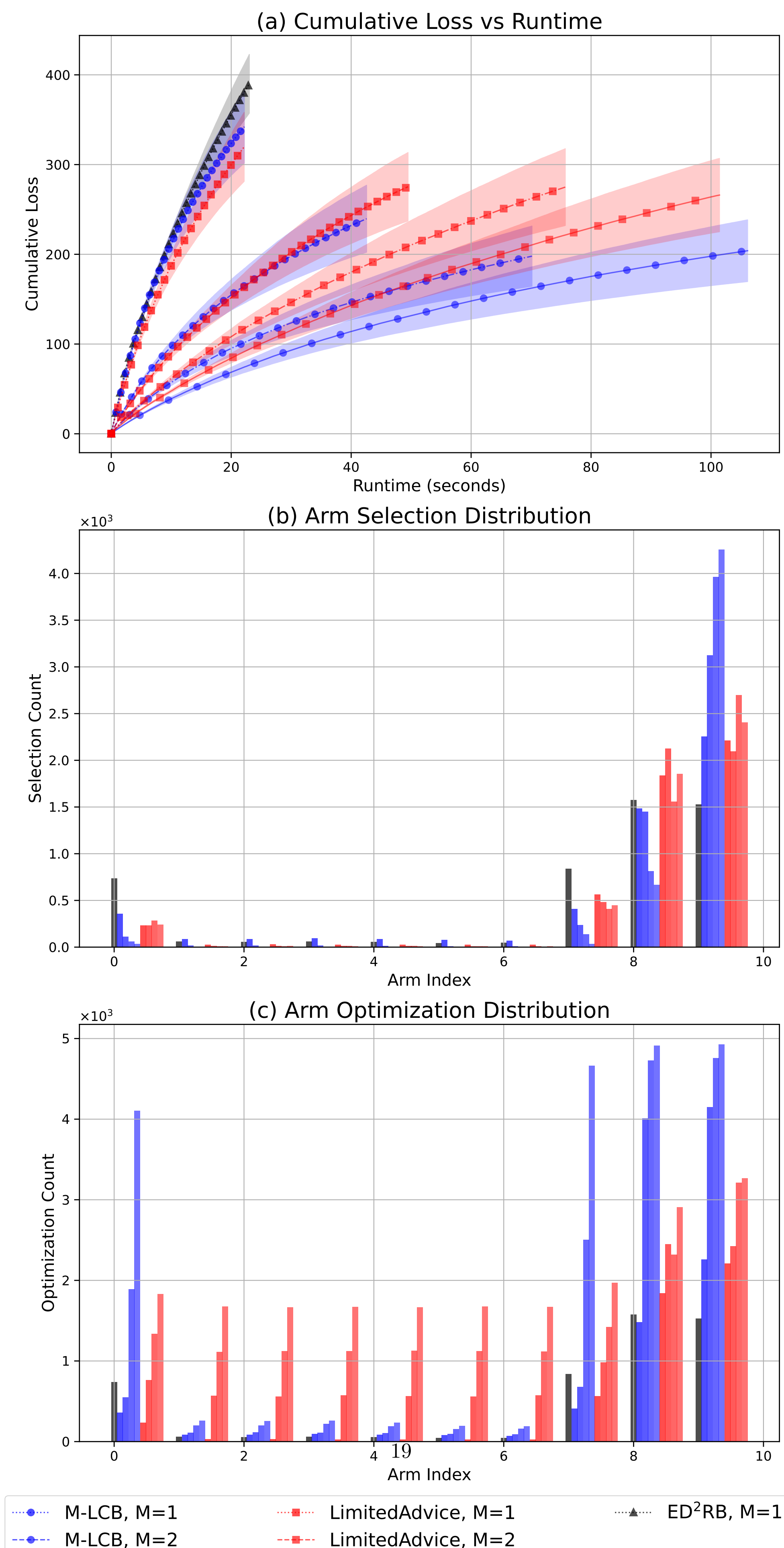
(a) Cumulative Loss vs Runtime
Cumulative Loss
Runtime (seconds)
0
100
200
300
400
0
20
40
60
80
100
(b) Arm Selection Distribution
×10^3
Selection Count
Arm Index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
2
4
6
8
10
(c) Arm Optimization Distribution
×10^3
Optimization Count
Arm Index
0
1
2
3
4
5
0
2
4
6
8
10
M-LCB, M=1
M-LCB, M=2
LimitedAdvice, M=1
LimitedAdvice, M=2
ED²RB, M=1

Contents

A Lower Bounds

We establish minimax lower bounds for our problem setup. The proof combines information-theoretic arguments (as in [Seldin et al.(2014)]) with the internal hardness of heavy-tailed bandits [Bubeck et al.(2013)]. The main idea is to construct a family of perturbed games, relate the probability of identifying the optimal expert to KL divergences via Pinsker's inequality, and transfer internal regret bounds from the null game (where all experts are identical) to the perturbed games.

Bandit setting and regret conventions. For concreteness, we consider experts represented by independent two–armed stochastic bandit problems. Each expert $h \in [K]$ has two arms with losses with expectations are $\ell_{h,1}$ and $\ell_{h,2}$, respectively. We note $\ell_h^\star = \min\{\mu_{h,1},\, \mu_{h,2}\}$ At each round $t = 1, \ldots, T$, expert h selects an arm $I_t \in \{1, 2\}$ and receives the corresponding loss $\ell_{h,I_t,t}$. The (expected) cumulative regret of expert h within its own subproblem is defined as

$$R_h^{\text{in}}(T) = \sum_{t=1}^{T} \mathbb{E}[\ell_{h,I_t}] - T \cdot \ell_h^\star,$$

where the subscript "in" emphasizes that this regret is *internal* to the metaprocedure. That is, each expert h acts as an independent learning agent whose own regret $R_h^{\text{in}}(T)$ contributes to the overall regret of the meta–learner.

A.1 Class of α–hard stochastic tasks

Theorem 4 (Lower bound)**.** *Consider K experts, horizon T, and a per-round budget M. Fix $\alpha \in [0, 1]$. There exists a class $\mathcal{F}_\alpha$ satisfying Def. 2, s.t. if each expert $k \in [K]$ solves a problem $f_k \in \mathcal{F}_\alpha$, then, for sufficiently small $\sqrt{\frac{K \log K}{MT}}$ and for any learning algorithm $\mathscr{A}_k$ and meta-procedure $\mathcal{P}$*

$$\sup_{f_k \in \mathcal{F}_\alpha,\, k \in [K]} \mathbb{E}\,\mathrm{Reg}(T) \geq c_1 \sqrt{\frac{K\,T}{M}} \;+\; c_2\, T^\alpha \left(\frac{K}{M}\right)^{1-\alpha},$$

where $c_1, c_2 > 0$ are absolute constants.

We consider heavy-tailed multi-armed bandits [Bubeck et al.(2013)] as base to build $\mathcal{F}_\alpha$. Each arm i provides rewards with mean μ_i and $(1 + \beta)$-moment bounded noise:

$$\mathbb{E}_{X \sim \nu_i} |X - \mu_i|^{1+\beta} \leq u, \tag{13}$$

for some $u > 0$ and $\beta \in (0, 1]$.

In our proof, as the canonical class $\mathcal{F}_\alpha$, we consider the *heavy-tailed multi-armed bandits* introduced by [Bubeck et al.(2013)]. From their analysis, the following corollary holds:

Corollary 1 (from [Bubeck et al.(2013)], Thm. 2)**.** *For a two-armed heavy-tailed bandit satisfying* (13)*, there exist distributions ν_1, ν_2 with $u = 1$ and gap $\ell_{h,1} - \ell_{h,2} = \Delta$ s.t., for any algorithm and any horizon n,*

$$R^{in}(n) \geq n\Delta\Big(1 - c_\beta \sqrt{n\Delta^{\frac{1+\beta}{\beta}}}\Big), \tag{14}$$

where $c_\beta > 0$ depends only on β. For fixed n, optimizing Δ as

$$\Delta = c_0\, n^{-\frac{\beta}{1+\beta}}$$

with sufficiently small $c_0 > 0$ yields

$$R^{in}(n) \geq c'\, n^{\frac{1}{1+\beta}}, \tag{15}$$

for some absolute constant $c' > 0$.

For heavy-tailed bandits this yields $\alpha = \frac{1}{1+\beta} \in [0.5, 1)$. The lower-bound argument below also applies to $\alpha \in [0, 0.5)$: in that regime the exploration term already dominates, and the internal-learning contribution can be lower bounded by zero without weakening the claimed rate order.

A.2 Construction of the composite game

Let K be the number of experts, M the per-round optimization budget, and T the horizon. We construct K perturbed games together with one symmetric *null game*. The setup depends on two small parameters: $\Delta > 0$ (internal hardness) and $\varepsilon > 0$ (cross-expert separation).

Perturbed games. In the h-th perturbed game, expert h faces a two-armed bandit with means $\ell_{h,1} = \frac{1}{2} - \frac{\varepsilon}{2}$ and $\ell_{h,2} = \frac{1}{2} - \frac{\varepsilon}{2} - \Delta$, while any expert $h' \neq h$ faces $\ell_{h,1} = \frac{1}{2} + \frac{\varepsilon}{2}$ and $\ell_{h,2} = \frac{1}{2} + \frac{\varepsilon}{2} + \Delta$.

Thus the h-th expert is uniquely optimal in game h.

Null game. All experts face identical subproblems with $\ell_{h,1} = \frac{1}{2} + \frac{\varepsilon}{2}$ and $\ell_{h,2} = \frac{1}{2} + \frac{\varepsilon}{2} + \Delta$. Each subproblem is hard, i.e. with internal regret characterized by Corollary 1.

A.3 Step 1: Regret decomposition

At each round t, the learner selects a subset $S_t \subseteq [K]$ of at most M experts to update, and then chooses one expert $H_t \in S_t$ for prediction. Then algorithm suffer (pseudo) loss $\ell_t^{H_t} \in \{\ell_{H_t,1}, \ell_{H_t,2}\}$. Define the empirical frequencies

$$\hat{q}_h = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\{H_t = h\}, \qquad J \sim \hat{q},$$

where J is a random variable representing the expert index sampled according to $\hat{q}$. Denote by $\mathbb{P}_h$ the law of J under the h-th game, and let $\mathbb{E}_h[\cdot]$ denote expectations in that game. Then

$$\mathbb{P}_h(J = h) \;=\; \mathbb{E}_h\left[\frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\{H_t = h\}\right].$$

Then the expected regret in game h can be bounded as follows.

$$\begin{aligned}
R_h(T) &:= \mathbb{E}_h\Bigg[\sum_{t=1}^{T} (\ell_t^{H_t} - (\frac{1}{2} - \frac{\varepsilon}{2} - \Delta))\Bigg] \\
&= \mathbb{E}_h\Bigg[\sum_{t=1}^{T}\Bigg[\mathbf{1}\{H_t = h\}\Big(\ell_t^{H_t} - (\frac{1}{2} - \frac{\varepsilon}{2} - \Delta)\Big) + \\
&\qquad\qquad + \mathbf{1}\{H_t \neq h\}\Big(\ell_t^{H_t} - (\frac{1}{2} - \frac{\varepsilon}{2} - \Delta)\Big)\Bigg]\Bigg] \geq \\
&= \varepsilon T \sum_{h' \neq h} \mathbb{P}_h(J = h') + \mathbb{E}_h\Bigg[\sum_{t=1}^{T}\Big(\ell_t^{H_t} - \ell^{\star}_{H_t}\Big)\Bigg] \\
&= \varepsilon T (1 - \mathbb{P}_h(J = h)) + \mathbb{E}_h\Bigg[\sum_{t=1}^{T}\Big(\ell_t^{H_t} - \ell^{\star}_{H_t}\Big)\Bigg],
\end{aligned}$$

The inequality is obtained using the add-subtract trick with $\varepsilon/2$ in the second term.

Taking the supremum over games gives

$$\mathrm{Reg}(T) \geq \sup_h R_h(T) \geq$$

$$\varepsilon T \left(1 - \frac{1}{K} \sum_{h=1}^{K} \mathbb{P}_h(J = h)\right) + \frac{1}{K} \sum_{h=1}^{K} \mathbb{E}_h \sum_{t=1}^{T} (\ell_t^{H_t} - \ell^{\star}_{H_t}). \tag{16}$$

We refer to the first term as the *identification term*, and to the second as the *internal regret term*, since grouping by arms reveals it as the sum of internal regrets of experts over their prediction rounds.

A.4 Step 2: Pinsker and null-game internal bound

Lemma 7 (Pinsker's inequality)**.** *For any h and event A,*

$$|\mathbb{P}_h(A) - \mathbb{P}_\emptyset(A)| \leq \sqrt{\tfrac{1}{2}\mathrm{KL}(\mathbb{P}_\emptyset \| \mathbb{P}_h)}.$$

In particular:

$$\mathbb{P}_h[J = h] \leq \mathbb{P}_\emptyset[J = h] + \sqrt{\tfrac{1}{2}\mathrm{KL}(\mathbb{P}_\emptyset \| \mathbb{P}_h)}$$

To bound the *identification term*, by the concavity of the square root we get:

$$\frac{1}{K}\sum_{h=1}^{K}\mathbb{P}_h[J=h] \le \frac{1}{K} + \sqrt{\frac{1}{2K}\sum_{h=1}^{K}\mathrm{KL}(\mathbb{P}_\emptyset\|\mathbb{P}_h)}. \tag{17}$$

To bound the *internal regret term* we form the following Lemma:

Lemma 8 (Internal regret in perturbed games). *Let $\alpha \in (0.5, 1]$ and suppose each expert's subproblem satisfies* (14). *Choose Δ as in* (19)*, i.e.*

$$\Delta = c_0\big(\tfrac{K}{MT}\big)^{1-\alpha}, \tag{18}$$

with c_0 sufficiently small. If the parameters K, M, T are s.t. $\sqrt{\frac{K\log(8K)}{MT}}$ is sufficiently small, and for a perturbed game h the divergence satisfies $\mathrm{KL}(P_\emptyset\|P_h) \le \frac{1}{2}$, then

$$\mathbb{E}_h\left[\sum_{t=1}^{T}\big(\ell_t^{H_t} - \ell^\star_{H_t}\big)\right] \ge c'' T^\alpha\left(\frac{K}{M}\right)^{1-\alpha},$$

for some constant $c'' > 0$ independent from M, K, T.

A.5 Step 3: KL computation for identification term

Lemma 9 (KL for T rounds). *For each $h \in [K]$,*

$$\mathrm{KL}(P_\emptyset\|P_h) \ \le\ \frac{36\,\max\{\varepsilon,\Delta\}^2}{1-9\max\{\varepsilon,\Delta\}^2}\,T.$$

Moreover,

$$\sum_{h=1}^{K}\mathrm{KL}(P_\emptyset\|P_h) \ \le\ \frac{36\,\max\{\varepsilon,\Delta\}^2}{1-9\max\{\varepsilon,\Delta\}^2}\,MT.$$

A.6 Step 4: Putting all together

From (16), the regret is lower bounded by the sum of an identification term and an internal-regret term. We use different parameter choices in the two regimes.

Case $\alpha \le 1/2$. Set $\Delta = \varepsilon$ and choose

$$\varepsilon = \gamma\sqrt{\tfrac{K}{MT}}$$

with sufficiently small $\gamma > 0$. Combining (17) with Lemma 9 gives

$$\sup_h R_h(T) \ge \varepsilon T\left(1 - \tfrac{1}{K} - c_{\mathrm{id}}\varepsilon\sqrt{\tfrac{MT}{K}}\right) \ge c_1\sqrt{\tfrac{KT}{M}},$$

where the last inequality holds for sufficiently small γ. Since

$$\frac{T^\alpha(K/M)^{1-\alpha}}{\sqrt{KT/M}} = \left(\frac{K}{MT}\right)^{1/2-\alpha} \le 1$$

under the stated smallness condition, the same lower bound implies

$$\sup_h R_h(T) \ge c_1'\left(\sqrt{\tfrac{KT}{M}} + T^\alpha\left(\tfrac{K}{M}\right)^{1-\alpha}\right)$$

for a smaller constant $c_1' > 0$.

Case $\alpha > 1/2$. Choose Δ as in (18) and set $\varepsilon = \gamma T^{-1/2}$ with $\gamma > 0$ sufficiently small. By Lemma 9, this choice ensures $\mathrm{KL}(P_\emptyset \| P_h) \le 1/2$ for every h, so Lemma 8 applies. Hence the internal-regret term in (16) satisfies

$$\frac{1}{K}\sum_{h=1}^{K} \mathbb{E}_h\left[\sum_{t=1}^{T}\left(\ell_t^{H_t} - \ell_{H_t}^{\star}\right)\right] \ge c_2\, T^{\alpha}\left(\tfrac{K}{M}\right)^{1-\alpha}.$$

Lower bounding the identification term by zero yields the same lower bound for $\mathrm{Reg}(T)$. Moreover,

$$\frac{\sqrt{KT/M}}{T^{\alpha}(K/M)^{1-\alpha}} = \left(\frac{K}{MT}\right)^{\alpha-1/2} \le 1$$

under the stated smallness condition. Therefore, after reducing constants,

$$\mathrm{Reg}(T) \ge c_2'\left(\sqrt{\tfrac{KT}{M}} + T^{\alpha}\left(\tfrac{K}{M}\right)^{1-\alpha}\right).$$

Combining the two cases proves Theorem 4. The constants may change between displays, but they remain independent of K, M, T and of the meta-procedure.

A.7 Proofs of Lemmas for Lower Bounds

Corollary 1. The construction follows the proof of Theorem 2 in [Bubeck et al.(2013)]. Let ν_1, ν_2 be the two heavy-tailed distributions defined therein, which satisfy the moment condition (13) with $u = 1$. By reduction to the Bernoulli case (see also [Bubeck(2010), Theorem 2.6]), the expected regret satisfies

$$R_n \ge n\Delta\Big(1 - \sqrt{n\mathrm{KL}(\nu_2\|\nu_1)}\Big).$$

Using the bound $\mathrm{KL}(\nu_2\|\nu_1) \le C_\beta \Delta^{\frac{1+\beta}{\beta}}$ for a constant $C_\beta > 0$ gives (14). Optimizing over Δ by setting $\Delta = c_0 n^{-\frac{\beta}{1+\beta}}$ and taking c_0 small enough makes the parenthesis positive, yielding (15). □

A.7.1 Zero Game analysis

Lemma 10 (Concentration of update counts)**.** *Consider null game. For any $\eta \in (0,1)$ there exists an absolute constant $C > 0$ s.t., with*

$$\delta = C\sqrt{\frac{K\log(2K/\eta)}{MT}} \in (0,1),$$

the following holds with probability at least $1-\eta$:

$$\left|n_i(T) - \tfrac{MT}{K}\right| \;\le\; \delta\,\tfrac{MT}{K} \qquad \textit{simultaneously for all } i \in [K].$$

Denote this event $\mathcal{E}_{conc}$.

Lemma 11 (Internal regret lower bound under concentration)**.** *Assume that for each expert $i \in [K]$, the internal subproblem satisfies Equation 14 for some constants $c_\beta > 0$, $\beta \in (0,1]$, and any $\Delta > 0$. Let $\alpha = \frac{1}{1+\beta}$ and let $\mathcal{E}_{\mathrm{conc}}$ denote the concentration event from Lemma 10. Then, on $\mathcal{E}_{\mathrm{conc}}$, choosing*

$$\Delta = c_0\left((1+\delta)\tfrac{MT}{K}\right)^{-(1-\alpha)}, \qquad c_0 \le \tfrac{1}{4c_\beta}, \tag{19}$$

we have

$$\sum_{t=1}^{T}\left(\ell_t^{H_t} - \ell_{H_t}^{\star}\right) \ge c' T^{\alpha}\left(\tfrac{K}{M}\right)^{1-\alpha}, \tag{20}$$

for some constant $c' > 0$ depending only on c_0 and α.

Lemma 10. We work under the probability distribution $\mathbb{P}_\emptyset$ induced by the randomization of the learner in the null game. To enforce symmetry even for deterministic algorithms, we assume that before the game begins, the K expert indices are randomly permuted. Hence, by symmetry, for every t and i,

$$p_t^{(i)} := \mathbb{E}_\emptyset[\mathbf{1}\{i \in S_t\} \mid \mathcal{F}_{t-1}] = \Pr_\emptyset(i \in S_t \mid \mathcal{F}_{t-1}) = \frac{M}{K},$$

and therefore $\sum_{t=1}^{T} p_t^{(i)} = MT/K$.

Define the martingale-difference sequence

$$X_t^{(i)} := \mathbf{1}\{i \in S_t\} - p_t^{(i)}, \quad \mathbb{E}_\emptyset[X_t^{(i)} \mid \mathcal{F}_{t-1}] = 0,$$

with $|X_t^{(i)}| \leq 1$. Then

$$n_i(T) - \tfrac{MT}{K} = \sum_{t=1}^{T} X_t^{(i)} =: S_T^{(i)}.$$

Let $V_T^{(i)}$ be the predictable quadratic variation:

$$\begin{aligned} V_T^{(i)} &= \sum_{t=1}^{T} \mathrm{Var}_\emptyset(\mathbf{1}\{i \in S_t\} \mid \mathcal{F}_{t-1}) \\ &= \sum_{t=1}^{T} p_t^{(i)}(1 - p_t^{(i)}) \leq \sum_{t=1}^{T} p_t^{(i)} = \tfrac{MT}{K}. \end{aligned}$$

Applying Freedman's inequality (martingale Bernstein bound), for any $u > 0$,

$$\mathbb{P}_\emptyset\Big(|S_T^{(i)}| \geq u\Big) \leq 2\exp\left(-\frac{u^2}{2(V_T^{(i)} + u/3)}\right).$$

Set $u = \delta \frac{MT}{K}$ with $\delta \in (0,1)$. Using $V_T^{(i)} \leq (M/K)T$ and $u \leq (MT/K)$ gives

$$\mathbb{P}_\emptyset\big(\big|n_i(T) - \tfrac{MT}{K}\big| \geq \delta \tfrac{MT}{K}\big) \;\leq\; 2\exp\big(-c\delta^2 \tfrac{M}{K} T\big)$$

for some absolute constant $c \in (0,1)$.

Finally, applying a union bound over all $i \in [K]$ yields

$$\mathbb{P}_\emptyset\Big(\max_i |n_i(T) - \tfrac{MT}{K}| \geq \delta \tfrac{MT}{K}\Big) \;\leq\; 2K\exp\big(-c\delta^2 \tfrac{M}{K} T\big)\,.$$

Choosing

$$\delta = C\sqrt{\tfrac{K\log(2K/\eta)}{MT}}$$

with sufficiently large C ensures that the right-hand side is at most η. Hence the stated event $\mathcal{E}_{\mathrm{conc}}$ holds with probability at least $1 - \eta$. □

Lemma 11. Summing internal regret across experts and substituting this into (14),

$$\sum_{t=1}^{T}\sum_{h\in S_t}(\ell_t^h - \ell_h^\star) = \sum_{h=1}^{K} R_h^{\mathrm{in}}\big(n_h(T)\big) \geq \Delta\sum_{h=1}^{K} n_h(T)\Big(1 - \sqrt{n_h(T)\Delta^{\frac{1+\beta}{\beta}}}\Big).$$

Since the played expert is uniform in S_t in the null game,

$$\sum_{t=1}^{T}(\ell_t^{H_t} - \ell_{H_t}^\star) = \frac{1}{M}\sum_{h=1}^{K} R_h^{\mathrm{in}}\big(n_h(T)\big) \geq \frac{1}{M}\Delta\sum_{h=1}^{K} n_h(T)\Big(1 - \sqrt{n_h(T)\Delta^{\frac{1+\beta}{\beta}}}\Big).$$

Under $\mathcal{E}_{\mathrm{conc}}$, all counts satisfy $n_h(T) \in [(1-\delta)\frac{MT}{K}, (1+\delta)\frac{MT}{K}]$ and $\sum_h n_h(T) = MT$, hence

$$\sum_{t=1}^{T}(\ell_t^{H_t} - \ell_{H_t}^\star) \geq \sum_{h=1}^{K} R_h^{\mathrm{in}}(n_h(T)) \geq T\Delta\left(1 - c_\beta\sqrt{(1+\delta)\tfrac{MT}{K}\Delta^{\frac{1+\beta}{\beta}}}\right).$$

Choosing Δ as in (19) ensures that $c_\beta\sqrt{(1+\delta)\frac{MT}{K}\Delta^{\frac{1+\beta}{\beta}}} \leq \frac{1}{2}$, hence

$$\sum_{t=1}^{T}(\ell_t^{H_t} - \ell_{H_t}^\star) \geq \tfrac{1}{2}T\Delta = c'T^\alpha\big(\tfrac{K}{M}\big)^{1-\alpha},$$

which yields (20). □

Lemma 8. Specify parameters for Lemma 10. Take $\eta = \frac{1}{4}$ and let K, M, T s.t. $\delta = C\sqrt{\frac{K\log(2K/\eta)}{MT}} \leq 1/2$. Then in zero game $\mathcal{E}_{\mathrm{conc}} = \{\forall\ h \in [K] : |n_h(T) - \frac{MT}{K}| \leq \frac{1}{2}\frac{MT}{K}\}$ is satisfied with probability $\geq \frac{3}{4}$.
Since $\mathrm{KL}(P_\emptyset \| P_h) \leq 1/2$, By Pinsker's inequality,

$$\left|\mathbb{P}_h(\mathcal{E}_{\mathrm{conc}}) - \mathbb{P}_\emptyset(\mathcal{E}_{\mathrm{conc}})\right| \leq \sqrt{\tfrac{1}{2}\mathrm{KL}(P_\emptyset \| P_h)} \leq \sqrt{\kappa/2} = 1/2,$$

hence $\mathbb{P}_h(\mathcal{E}_{\mathrm{conc}}) \geq 1 - 1/4 - 1/2 = 1/4$. On $\mathcal{E}_{\mathrm{conc}}$, Lemma 11 yields the realized regret bound

$$\sum_{t=1}^{T}(\ell_t^{H_t} - \ell_{H_t}^\star) \ \geq\ c' T^\alpha \big(\tfrac{K}{M}\big)^{1-\alpha}.$$

Taking expectations under $\mathbb{P}_h$ gives

$$\mathbb{E}_h\left[\sum_{t=1}^{T}(\ell_t^{H_t} - \ell_{H_t}^\star)\right] \geq$$

$$\geq\ c' T^\alpha \big(\tfrac{K}{M}\big)^{1-\alpha}\, \mathbb{P}_h(\mathcal{E}_{\mathrm{conc}}) \geq (1/4) c' T^\alpha \big(\tfrac{K}{M}\big)^{1-\alpha}.$$

And constants adsorm into c'' □

A.7.2 KL computation

Lemma 9. The proof follows [Seldin et al.(2014)], and provided here for completeness. The only change is the KL bounding, since in our setup on each arm the not a fixed Bernoulli distribution is specified, but a mixture of distributions. By the data-processing inequality, $\mathrm{KL}(P_\emptyset \| P_h) \leq \mathrm{KL}(\tilde{P}_\emptyset^T \| \tilde{P}_h^T)$, so it suffices to bound the latter. Using the chain rule for KL divergence,

$$\begin{aligned}
&\mathrm{KL}(\tilde{\mathbb{P}}_\emptyset^T \| \tilde{\mathbb{P}}_h^T) \\
&= \sum_{t=1}^{T}\sum_{o_1^{t-1}} \tilde{\mathbb{P}}_\emptyset^{t-1}(o_1^{t-1}) \mathrm{KL}\Big(\tilde{\mathbb{P}}_\emptyset^t(\cdot \mid o_1^{t-1}) \Big\| \tilde{\mathbb{P}}_h^t(\cdot \mid o_1^{t-1})\Big) \\
&= \sum_{t=1}^{T}\sum_{o_1^{t-1}} \tilde{\mathbb{P}}_\emptyset^{t-1}(o_1^{t-1}) \mathbf{1}\{h \in O_t \mid o_1^{t-1}\} \times \\
&\times \mathrm{KL}\Big(\tilde{\mathbb{P}}_\emptyset^t(\cdot \mid o_1^{t-1}) \Big\| \tilde{\mathbb{P}}_h^t(\cdot \mid o_1^{t-1})\Big) \\
&\leq \frac{6\varepsilon^2}{1-\varepsilon^2} E_\emptyset\left[\sum_{t=1}^{T} \mathbf{1}\{h \in O_t\}\right]
\end{aligned}$$

The Inequality is from fact, that each arm at moment t has a bernoulli distribution. Denote $\varepsilon' = \max\{\varepsilon, \Delta\}$ h in h game is with parameter $p_1 \in \left[\frac{1}{2} + \frac{\varepsilon'}{2},\ \frac{1}{2} + \frac{3\varepsilon'}{2}\right]$. And h in zero game with parameter $p \in \left[\frac{1}{2} - \frac{3\varepsilon'}{2},\ \frac{1}{2} - \frac{\varepsilon'}{2}\right]$. Then, by the standard quadratic upper bound on Bernoulli KL divergence,

$$\mathrm{KL}(\mathrm{Ber}(p) \| \mathrm{Ber}(p_1)) \leq \frac{(p-p_1)^2}{p_1(1-p_1)} \leq \frac{36\varepsilon'^2}{1-9\varepsilon'^2}.$$

To obtain the total bound, we sum over $h \in [K]$:

$$\sum_{h=1}^{K} \mathrm{KL}(P_\emptyset \| P_h) \leq \frac{36\varepsilon'^2}{1-9\varepsilon'^2} \mathbb{E}_\emptyset\left[\sum_{t=1}^{T}\sum_{h=1}^{K} \mathbf{1}\{h \in O_t\}\right].$$

The first inequality then follows from the fact, that each of the arm is selected no more than T times. At each round t, at most M experts are observed, i.e. $\sum_{h=1}^{K} \mathbf{1}\{h \in O_t\} \leq M$. Hence

$$\sum_{h=1}^{K} \mathrm{KL}(P_\emptyset \| P_h) \leq \frac{36\varepsilon'^2}{1-9\varepsilon'^2} MT,$$

which completes the proof. □